

Backtest Summary: A Tale of Two Strategies

Comparative Performance Analysis | `MACD\_NM\_MTF` vs. `GLSRDemo\_2`

Strategy	Trades	Avg Profit %	Tot Profit USDT	Tot Profit %	Avg Duration	Win	Draw	Loss	Win%	Drawdown
`MACD_NM_MTF_Strategy`	34	4.05	2223.469	222.35	7 days, 19:07:08	20	0	14	58.8	13.86%
`GLSRDemo_2`	68	2.33	3188.719	318.87	3 days, 18:56:08	33	0	35	48.5	21.14%

The Trade-Off

`GLSRDemo\_2` delivers a higher `Total Profit %` (318.87% vs. 222.35%) but at the cost of significantly higher `Drawdown` (21.14% vs. 13.86%).

Efficiency vs. Activity

`MACD\_NM\_MTF` appears more efficient, with a much higher `Avg Profit %` per trade (4.05% vs. 2.33%) and a higher `Win%` (58.8% vs. 48.5%) from half the number of trades (34 vs. 68).

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Pacing

Note the `Avg Duration`. `MACD\_NM\_MTF` holds trades for over a week on average, while `GLSRDemo\_2` is more active with a duration under 4 days.

Strategy 1 Logic: `MACD\_NM\_MTF`

A Classic Trend-Following Approach with MACD and Moving Average Filters

This section details the entry and exit conditions for the strategy. The logic aims to capture sustained market trends by aligning price action with multiple momentum indicators.

Entry Logic

LONG ENTRY

```
// Enter when price crosses above the main moving average,  
// with confirmation from a secondary MA order and positive MACD momentum.  
if (  
    close crosses over ma_main    AND    // Primary trend signal  
    ma_order == 1                AND    // Confirmatory trend alignment  
    macd_hist > 0                // Momentum confirmation  
) then ENTER LONG;
```

SHORT ENTRY

```
// Enter when price crosses below the main moving average,  
// with confirmation from a secondary MA order and negative MACD momentum.  
if (  
    close crosses under ma_main    AND    // Primary trend signal  
    ma_order == -1                AND    // Confirmatory trend alignment  
    macd_hist < 0                // Momentum confirmation  
) then ENTER SHORT;
```

Exit Logic & Visual Explanation

The strategy utilizes a standard combination of profit-taking and risk-management exits:

- **"Take Profit (TP):** A pre-defined price target to lock in gains.
- **"Stop Loss (SL):** A pre-defined price level to cap losses.
- **"Trailing Stop Loss (Trailing_SL):** A dynamic stop loss that trails the price to protect accrued profits as a trade moves in a favorable direction.



Strategy 1 Performance Metrics: The Anatomy of a Conservative System

A Deep Dive into the `MACD\_NM\_MTF` Backtest from 2024-01-03 to 2025-11-19

Metric	Value
Backtesting from	2024-01-03 02:00:00
Backtesting to	2025-11-19 20:00:00
Trading Mode	Isolated Futures
Max open trades	1
Total/Daily Avg Trades	34 / 0.05
Starting balance	1000 USDT
Final balance	3223.469 USDT
Absolute profit	2223.469 USDT
Total profit %	222.35%
CAGR %	86.41%
Sortino	1.16
Sharpe	0.31
Calmar	44.67
SQN	1.86
Profit factor	2.23
Expectancy (Ratio)	65.40 (0.51)
Avg. daily profit	3.241 USDT
Avg. stake amount	1679.691 USDT
Total trade volume	113605.391 USDT
Long / Short trades	14 / 20
Long / Short profit %	73.43% / 148.91%
Best Pair	ETH/USDT:USDT 222.35%
Worst Pair	ETH/USDT:USDT 222.35%
Best trade	ETH/USDT:USDT 25.44%
Worst trade	ETH/USDT:USDT -7.87%
Min/Max/Avg. Duration Winners	1d 12:00 / 18d 16:00 / 7d 12:27
Min/Max/Avg. Duration Losers	0d 22:00 / 22d 02:00 / 8d 04:39
Max Consecutive Wins / Loss	5 / 2
Min balance	928.578 USDT
Max balance	3223.469 USDT
Max % of account underwater	13.86%
Absolute drawdown	403.841 USDT (13.86%)

Solid Profitability

A 'Profit Factor' of **2.23** indicates that gross profits were more than double the gross losses.

Strong Downside Protection

The 'Sortino Ratio' of **1.16** is a strong indicator of good risk-adjusted returns, specifically considering downside volatility.

Psychological Stability

The 'Max Consecutive Wins / Loss' of **5 / 2** is a standout metric. The ability to weather losing streaks is excellent, making it an easier strategy to follow.

Contained Risk

'Max % of account underwater' is only **13.86%**, confirming the strategy's conservative nature seen on the first slide.

Market Bias

The strategy performed significantly better on the short side ('148.91%' profit) than the long side ('73.43%' profit) during this specific backtest period.

Freqtrade Profit plot



Strategy 2 Logic: `GLSRDemo\_2`

A Relative Value Approach Using Ratio Bollinger Bands and Open Interest

This strategy employs a mean-reversion methodology on a ratio of assets, filtered by market participation (Open Interest). It aims to capitalize on statistical deviations from the norm.

Entry Logic

LONG ENTRY

```
// Enter when a ratio of two assets becomes "oversold" (crosses below its lower Bollinger Band),  
// but only when market participation is high, confirming the signal's validity.  
if (  
    gls_ratio crosses under gls_ratio_bbands_d1(len) AND // Mean-reversion signal  
    oi > oi_bband_d1(len) AND // High market participation filter  
    macd_hist > 0 // Momentum confirmation  
) then ENTER LONG;
```

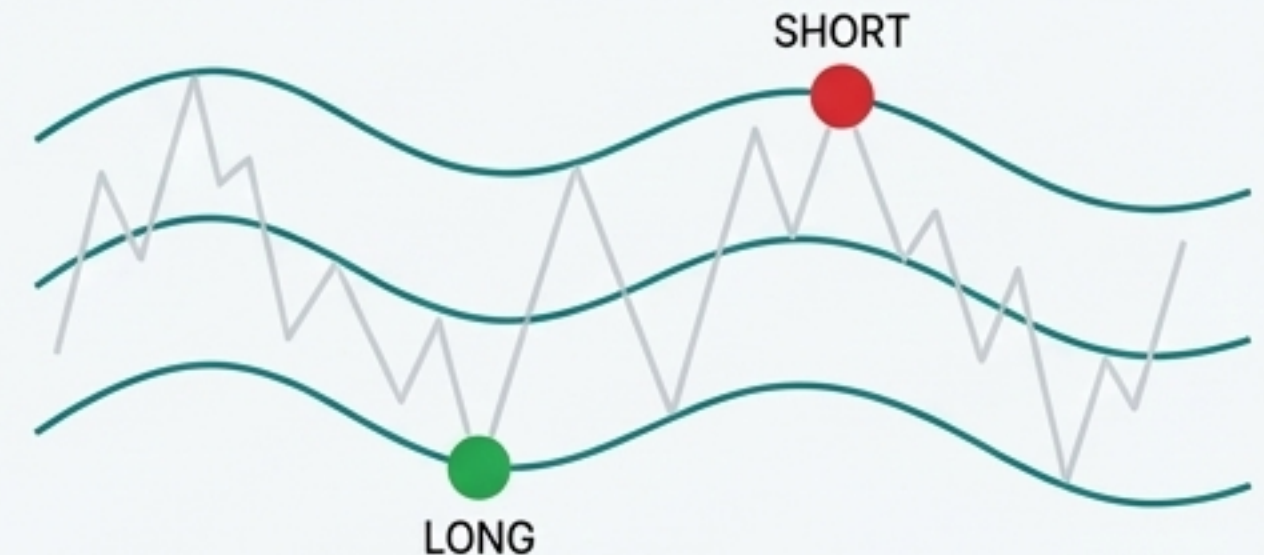
SHORT ENTRY

```
// Enter when the ratio becomes "overbought" (crosses above its upper Bollinger Band),  
// again confirmed by high Open Interest.  
if (  
    gls_ratio crosses over gls_ratio_bbands_u1(len) AND // Mean-reversion signal  
    oi > oi_bbands_d1(len) AND // High market participation filter  
    macd_hist < 0 // Momentum confirmation  
) then ENTER SHORT;
```

Exit Logic

Exits are triggered by a reversal signal or standard risk management parameters:

- **Opposite Signal:** An entry signal in the opposite direction will close the current trade.
- **Stop Loss (SL):** A pre-defined price level to cap losses.
- **Take Profit (TP):** A pre-defined price target to lock in gains.



Strategy 2 Performance Metrics: The Profile of an Aggressive System

A Deep Dive into the `GLSRDemo\_2` Backtest from 2024-01-03 to 2025-11-19

Metric	Value
Backtesting from	2024-01-03 02:00:00
Backtesting to	2025-11-19 20:00:00
Trading Mode	Isolated Futures
Max open trades	1
Total/Daily Avg Trades	68 / 0.1
Starting balance	1000 USDT
Final balance	4108.719 USDT
Absolute profit	3108.719 USDT
Total profit %	318.87%
CAGR %	112.10%
Sortino	2.04
Sharpe	0.61
Calmar	40.95
SQN	2.63
Profit factor	2.23
Expectancy (Ratio)	45.72 (0.63)
Avg. daily profit	4.532 USDT
Avg. stake amount	1930.289 USDT
Total trade volume	262694.164 USDT
Long / Short trades	37 / 31
Long / Short profit %	152.70% / 158.17%
Best Pair	ETH/USDT:USDT 310.87%
Worst Pair	ETH/USDT:USDT 310.87%
Best trade	ETH/USDT:USDT 10.38%
Worst trade	ETH/USDT:USDT -4.31%
Min/Max/Avg. Duration Winners	0d 00:00 / 12d 14:00 / 4d 17:11
Min/Max/Avg. Duration Losers	0d 03:00 / 9d 12:00 / 2d 06:26
Max Consecutive Wins / Loss	5 / 7
Min balance	938.886 USDT
Max balance	4108.719 USDT
Max % of account underwater	21.14%
Absolute drawdown	442.072 USDT (21.14%)

Superior Returns

The 'CAGR' of **112.10%** and 'Absolute profit %' of **318.87%** are significantly higher than Strategy 1.

Excellent Risk-Adjusted Performance

The 'Sortino Ratio' of **2.04** is outstanding and nearly double that of Strategy 1, indicating highly efficient returns relative to downside risk. The '**Sharpe Ratio**' (0.61) is also double.

The Price of Performance

'**Max % of account underwater**' is a substantial **21.14%**. This volatility is a key characteristic.

Psychological Challenge

A 'Max Consecutive Loss' streak of **7** trades is a critical risk metric. This is over 3x longer than Strategy 1's losing streak and represents a significant test of a trader's discipline.

Win Rate

The Win % is below 50% at **48.5%**, typical for mean-reversion strategies that often aim for larger wins to offset more frequent small losses.

Freqtrade Profit plot



The Verdict: Choosing a Strategy Based on Trader Profile

A Side-by-Side Comparison of `MACD\_NM\_MTF` and `GLSRDemo\_2`

A direct comparison reveals two distinct profiles, each suited for a different type of trader.

Metric	MACD_NM_MTF (Strategy 1) Steady & Conservative	GLSRDemo_2 (Strategy 2) Aggressive & High-Frequency	Key Takeaway
Total Profit %	222.35%	318.87%	GLSRDemo_2 generates significantly higher absolute returns.
CAGR %	86.41%	112.10%	Superior compounded growth from `GLSRDemo_2`.
Max Drawdown %	13.86%	21.14%	MACD_NM_MTF offers far better capital preservation.
Sortino Ratio	1.16	2.04	GLSRDemo_2 provides superior returns for the downside risk taken.
Win %	58.8%	48.5%	MACD_NM_MTF has a higher probability of success on any given trade.
Total Trades	34	68	GLSRDemo_2 is twice as active, incurring more transaction costs.
Max Consec. Losses	2	7	MACD_NM_MTF is psychologically much easier to trade.

Conclusion

MACD_NM_MTF

Ideal for the **risk-averse trader** or those seeking a stable portfolio component. It prioritizes capital preservation and consistency, offering steady growth with minimal losing streaks.

GLSRDemo_2

Suited for the **aggressive trader** comfortable with higher volatility. It delivers superior returns and is more efficient from a risk-adjusted perspective (Sortino/Sharpe), but demands tolerance for larger drawdowns and longer periods of underperformance.